Chris Avanessian

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EDUCATION

Pacific Northwest National Laboratory — Richland, WA
Postdoctoral Research Associate — July 2026 – Present
Advisor: Dr. Niri Govind

Johns Hopkins University — Baltimore, MD
Ph.D. in Chemistry, Department of Chemistry — May 2026
Advisors: Professor David R. Yarkony (deceased 2024)
Professor Lan Cheng
Dissertation: “Photodetachment and Light–Matter Interactions Influenced by Conical Intersections.”

Johns Hopkins University — Baltimore, MD
M.A. in Chemistry, Department of Chemistry — May 2025

Union College — Schenectady, NY
B.S. in Chemistry (ACS accredited) and Physics with Honors — June 2021
Valedictorian
Senior thesis: “Microstructural Characterization of Catalytic Copper- and Chromium-Containing Alumina Aerogels.”

AWARDS, RECOGNITIONS, AND CERTIFICATIONS

2025 Johns Hopkins Teaching Academy
2021 ACS Undergraduate Award in Physical Chemistry
2021 Robert M. Fuller (1863) Senior Prize for outstanding work in chemistry
2021 Professor Frank Titus Memorial Prize in Physics
2021 Michael E. Hagerman Award for Student Mentoring and Professional Development
2020 Student Research Grant for senior thesis work
2020 Union College Research Fellowship for summer research
2020 Charles B. Hurd Prize to an outstanding student pursuing coursework in physical chemistry
2020 Mortimer T. Harvey (1917) Scholarship
2019 Student Conference Travel Grant
2019 Laudise Research Fellowship for summer research
2019 Robert M. Fuller (1863) Sophomore Prize for outstanding work in chemistry
2018 Chemistry Department First-Year Student Prize
2018 Wessel Ten Broeck Van Orden (1839) Prize for excellence in English composition
2017 Goldman Scholars - David Viniar (1976) Scholarship

RESEARCH CONTRIBUTIONS

Refereed Journal Publications

8. "Intersystem crossing in the molecular channel of ammonia photodissociation manipulated by an external laser field." **C. Avanessian**, *J. Chem. Phys*. **2025**, *163* (8), 084311. doi.org/10.1063/5.0284970

7. "Chromia-alumina catalytic aerogels prepared via rapid supercritical extraction." F. T. Fitzgerald, **C. Avanessian**, J. P. Santos, A. M. Anderson, B. A. Bruno, and M. K. Carroll, *J. Sol-Gel Sci. Technol.* **2025**, *116*, 2447–2459. doi.org/10.1007/s10971-025-06686-9

6. "Floquet-Engineered Photodissociation Simulated Using Coupled Potential Energy and Dipole Matrices." **C. Avanessian**, Y. Wang, and D. R. Yarkony, *J. Phys. Chem. Lett.* **2024**, *15* (39), 9905–9911. doi.org/10.1021/acs.jpclett.4c02312

5. "The anion photoelectron spectrum and diabatization of tetrazolyl." **C. Avanessian** and D. R. Yarkony, *J. Chem. Phys*. **2024**, *160* (21), 214312. doi.org/10.1063/5.0214635

4. "Effect of slurry processing on the properties of catalytically active copper-alumina aerogel material for applications in three-way catalysis." A. M. Anderson, B. A. Bruno, J. Santos, **C. Avanessian**, and M. K. Carroll, *J. Sol-Gel Sci. Technol*. **2022**, *102* (2), 422–436. doi.org/10.1007/s10971-022-05757-5

3. "Synthesis and characterization of plasmonic peptoid nanosheets." E. J. Robertson, **C. Avanessian**, J. R. Davis, A. K. Mahony, and E. V. Whitney, *Chem. Comm*. **2021**, *57* (22), 2748–2751. doi.org/10.1039/d1cc00092f

2. "Effect of Copper Loading in Copper-Alumina Aerogels on Three-Way Catalytic Performance." A. M. Anderson, B. A. Bruno, F. Dilone, M. T. LaRosa, T. F. Andre, **C. Avanessian**, and M. K. Carroll, *Emiss. Control Sci. Technol*. **2020**, *6* (3), 324–335. doi.org/10.1007/s40825-020-00165-z

1. "Analysis and Characterization of Etched Silica Aerogels." A. M. Stanec, A. M. Anderson, **C. Avanessian**, and M. K. Carroll, *J. Sol-Gel Sci. Technol*. **2020**, *94* (2), 406–415. doi.org/10.1007/s10971-020-05256-5

Oral Presentations

6. "Gold nanoparticle-embedded peptoid nanosheets: A new class of solution-based two-dimensional plasmonic materials." E. J. Robertson, W. O'Connor, **C. Avanessian**, J. Davis, A. K. Mahony, and E. V. Whitney. *ACS Spring 2022 National Meeting in San Diego, CA. Mar. 20, 2022.* Abstract

5. "Microstructural characterization of catalytic copper- and chromium-containing alumina aerogels." **C. Avanessian**, M. K. Carroll, and A. M. Anderson. *31st Steinmetz Symposium, Union College, held virtually. May 22, 2021.* Open access full video recording

4. "Spectroscopic and Microscopic Characterization of Gold-Nanoparticle-Containing Peptoid Monolayer Sheets." **C. Avanessian** and E. J. Robertson. *30th Steinmetz Symposium, Union College, held virtually. May 22, 2020.* Abstract

3. “Catalytic Copper-Alumina Aerogels: Synthesis, Elemental Mapping and Catalytic Activity.” **C. Avanessian**, F. Dilone, M. K. Carroll, A. M. Anderson, and B. A. Bruno. *30th Steinmetz Symposium, Union College, held virtually. May 22, 2020.* Abstract

2. “Peptoid monolayers as versatile templates for the two-dimensional assembly of hydrophobically modified gold nanoparticle at fluid interfaces.” E. J. Robertson, E. V. Whitney, A. K. Mahony, **C. Avanessian**, and J. Davis. *259th ACS National Meeting held virtually. Mar. 2020.* doi.org/10.1021/scimeetings.0c04036

1. “Synthesis and Characterization of Copper-Alumina Aerogels.” **C. Avanessian**, F. Dilone, M. K. Carroll, and A. M. Anderson. *New York Six Undergraduate Research Conference in Schenectady, NY. Oct. 26, 2019.*

Poster Presentations

8. “Extending laser cooling to non-linear molecules.” Z. Guo, **C. Avanessian**, and L. Cheng. *18th International Congress on Quantum Chemistry in Berkeley, CA. June 1, 2026.*

7. “Internal Conversion and Intersystem Crossing in Ammonia Photodissociation Manipulated by a Light-Induced Conical Intersection.” **C. Avanessian**. *29th Dynamics of Molecular Collisions Conference in Snowbird, UT. July 7, 2025.* Abstract

6. “The effect of a laser field on ammonia photodissociation simulated on accurate potential energy surfaces.” **C. Avanessian**, Y. Wang, and D. R. Yarkony. *2024 Molecular Interactions and Dynamics Gordon Research Conference and Seminar in Easton, MA. July 8, 2024.*

5. “Simulating the Anion Photoelectron Spectrum of Tetrazolyl with the Time-Independent Multimode Vibronic Coupling Approach.” **C. Avanessian** and D. R. Yarkony. *28th Dynamics of Molecular Collisions Conference in Snowbird, UT. July 12, 2023.* Abstract

4. “Microstructural characterization of catalytic copper- and chromium-containing alumina aerogels.” **C. Avanessian**, M. K. Carroll, and A. M. Anderson. *ACS Spring 2021 National Meeting held virtually. April 21, 2021.* Full poster

3. “Microscopic Characterization and Elemental Mapping of Catalytic Aerogels.” **C. Avanessian**, M. K. Carroll, and A. M. Anderson. *ACS Northeast Regional Meeting in Saratoga Springs, NY. June 23, 2019.*

2. “Microscopic Characterization and Elemental Mapping of Catalytic Aerogels.” **C. Avanessian** and M. K. Carroll. *29th Steinmetz Symposium, Union College, Schenectady, NY. May 10, 2019.* Abstract

1. “Calculating Surface Tension with Capillary Waves.” **C. Avanessian** and J. Wagner. *29th Steinmetz Symposium, Union College, Schenectady, NY. May 10, 2019.* Abstract

TEACHING AND RESEARCH EXPERIENCE

Substitute Lecturer
Johns Hopkins University

Spring 2026 Introductory Chemistry II

Teaching Apprenticeship
Johns Hopkins University

Summer 2025 Introductory Chemistry I
Supervisor: Professor Sunita Thyagarajan

Teaching Assistant
Johns Hopkins University

Spring 2023 Introductory Chemistry II

Fall 2022 Applied Chemical Equilibrium Lab

Spring 2022 Introductory Chemistry II

Fall 2021 Physical Chemistry I

Undergraduate Research Experience
Union College

Sep 2018 – June 2021 **Research in materials chemistry, Union College**
Advisor: Professor Mary K. Carroll
Project: "Microscopic Characterization and Elemental Mapping of Catalytic Aerogels."

Sep 2019 – June 2020 **Research in physical chemistry, Union College**
Advisor: Professor Ellen J. Robertson
Project: "Liquid-Liquid Interfacial Assembly of Peptoid Sheets."

Jan 2018 – June 2018 **Student-initiated physics research project, Union College**
Advisor: Professor Jef Wagner
Project: "Calculating Surface Tension with Capillary Waves."

Sep 2017 – June 2018 **Psychology research practicum, Union College**
Advisor: Professor Cay Anderson-Hanley
Project: "Interactive Physical and Cognitive Exercise Study."

WORKSHOPS

Johns Hopkins Teaching Institute
Teaching Academy, Center for Teaching Excellence and Innovation
Johns Hopkins University, Baltimore, MD, May 28-30, 2025.

OUTREACH

Johns Hopkins University, ChemDNA Graduate Student Mentor
Provided information and guidance on being a first-year PhD student in the Chemistry Department.

Fall 2025 and Spring 2026	Zhihan Zhao
Fall 2024 and Spring 2025	William Barr
Fall 2023 and Spring 2024	Minghao Wang
Fall 2022 and Spring 2023	Xiuze Wang

Johns Hopkins University, Undergraduate Student
Antonella Macoretta, rising second-year undergraduate student. Introduced the research of the Yarkony group and the theory that informs it. (Summer 2022)

TECHNICAL PROFICIENCY

Computational
Python (NumPy, Pandas, scikit-learn, Matplotlib) | Jupyter Notebook | Machine learning | Fortran | Git | GitHub | Bash | Linux | High-performance computing

Scientific programs
Gaussian | ORCA | MOLPRO | CFOUR | COLUMBUS | SHARC | Avogadro | Jmol | MOLDEN

Experimental
Differential scanning calorimetry | Scanning electron microscopy | Energy-dispersive X-ray spectroscopy | Atomic force microscopy | Powder X-ray diffraction | Optical microscopy | Infrared spectroscopy | UV-vis spectroscopy | Nuclear magnetic resonance spectroscopy

ACTIVITIES AND SERVICE

Maryland American Chemical Society section, executive committee officer	2026 – present
Johns Hopkins Chemistry Student Safety Committee, Safety Captain	2022 – 2026
Phi Beta Kappa, member	2020 – present
Sigma Pi Sigma, Legacy Member	2020 – present
Sigma Xi, Full Member	2019 – present
American Physical Society, member	2019 – present
American Chemical Society, member	2019 – present
Union College Gaming4U, secretary	2019 – 2021
Union College Chemistry and Biochemistry Club, secretary	2018 – 2021